SAFE TEST PLAN · MONITOR V2.0

Bringing the D1 monitor into service, one safe step at a time

The monitor only listens: it has no way to send the D1 anything, so **no step here can change what the D1 does**. The risks are the usual ones of working near batteries, and a wiring mistake. Each stage checks one thing before the next. Stop at the first thing that looks wrong.

At every step the Cerbo sends nothing: it can't change what the D1 does



What protects you

- The D1 protects itself (D1 manual). It stops if the 12 V side goes over 16–18 V or under 8 V, the 48 V side goes over 58–60 V or under 24 V, the current goes over 213 A, there's a short, or it gets hotter than 248 °F (120 °C) inside
- Your BMS and fuses on HV+, LV+ and KL30
- The cable only carries data, and the monitor never transmits on it

Before you start

- Fuses on HV+ and LV+ (D1 manual), and 3 A on KL30
- A multimeter, plus a DC clamp meter if you have one
- The D1 switched off (KL30/KL15) while you wire
- The Installation Guide (`docs/installation-guide.pdf`) at hand

Stage	What	Risk
1	Simulation on the Cerbo	None: no converter, no CAN bus
2	Wiring check, everything off	None: no power
3	Listen	None: the Cerbo sends nothing
4	Installed and watching for a day	None: the Cerbo sends nothing

1 Simulation: no converter at all

In Cerbo Setup, choose **9**. The real monitor runs on your Cerbo against a pretend D1, and no CAN socket is ever opened. A real installation keeps running beside it.

Check	OK
"DC-DC D1 (SIMULATED)" cards appear in the Cerbo's switch pane, and in VRM's Switches	
Its web page opens at **http://<Cerbo IP>:8089**, and the charts start filling after a minute	
The pretend 12 V side drains to 12.5 V, then Status shows **Autonomous** and it charges back to about 13.4 V (12 V side current +, 48 V side −)	
Every three minutes a pretend alternator pushes the 12 V side above 13.5 V and the pretend D1 boosts (12 V side current −, 48 V side +) until 12.3 V	

Stop with Ctrl+C. The simulated cards disappear. The pretend D1 follows the manual's logic, but it can't tell you anything about the real one.

2 Wiring check, everything off

Check	OK
Continuity: RJ45 pin 7 to white/brown, pin 8 to brown (guide step 2b)	
No continuity (no beep) between white/brown or brown and LV+, HV+ or GND	
Lever connector A holds the D1's CAN_H wire, white/brown and one resistor leg; B holds CAN_L, brown and the other leg. Every wire fully in, levers closed, resistor legs insulated. White/green and the other unused wires insulated (no ground wire)	
RJ45 not in the Cerbo: white/brown to brown reads about 120 Ω (your resistor at the D1)	
RJ45 in the Cerbo, with the VE.Can terminator in the other socket: about 60 Ω	

3 Listen

Power the D1 as normal (KL30 and KL15 on). In Cerbo Setup choose **2** (find), then **3** (watch).

Check	What you saw	OK
Converter found, at the right speed (if it reports bus errors, switch speed as the guide says)		
12 V side reading within about 0.2 V of your meter on the D1's LV+ to GND		
48 V side reading within about 0.5 V of your meter on HV+ to GND		
While the D1 is converting: which side shows + current? Check with the clamp meter which way the current really flows		
No alarms, or only alarms that make sense (for example low voltage when the battery really is low)		

4 Installed and watching

Choose **4** (install). Leave it running for a day.

Check	OK
Three cards on the Cerbo, in VRM and on the web page, with sensible readings	
The D1's temperature shows on the Overview's Alternator tile, as a **DC-DC D1** gauge on Levels › Environment and on the web page (OK). While it works hard, note the highest reading: up to 212 °F is fine, a High Temperature warning comes at 212 °F, and the D1 starts derating at 221 °F. Max seen: _____ °F	
The web page's charts cover the day, with no gaps you can't explain (a gap means the D1 went quiet, or the Cerbo restarted)	
The D1 behaves exactly as before: buck when the 12 V side is low, boost when it's high, standby otherwise	
After a Cerbo reboot the monitor comes back by itself (option 5 shows "Converter online")	

> **Please share what you found:** which way the current sign went in stage 3, and anything that didn't match. Open an issue on the project's GitHub page.

Tested and signed off by: _____ Date: _____